

A Sunflower Classification of Gaps in Maximal Sidon Spectra of Four-Prime Divisor Sets

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Abstract

For a finite set X of integers, let $\text{Spec}_{\max}(X)$ be the set of cardinalities of its inclusion-maximal Sidon subsets. We classify the possible gap mechanisms for squarefree divisor sets $n = pqrs$, $p < q < r < s$. Collision-transversal duality and a degree-two interval theorem first force a divisor contained in at least three minimal collision supports. An exact source reduction treats every such concurrence and, for odd p , reaches a final interface of 120 exact branch ideals. These are identified injectively with 120 identities in a canonical 897-branch reference ledger. A fail-closed final composition accounts for all 120 branches and leaves exactly six mechanisms—four polynomial families and two branches on a Pell-type conic. Their generic branch systems are the six arithmetic orientations of one s -centered three-petal sunflower, and their parameter formulas follow by elementary elimination. The even case is closed independently by an exact $891 + 6 = 897$ replay. No prime cutoff, floating-point decision, probabilistic primality assumption, or sunflower hypothesis is used in the reduction. Together with the companion ternary contiguity theorem, this makes four prime factors the first squarefree level at which a primitive gap can occur, while separated-prime lifting propagates such gaps to every higher squarefree level.

Keywords: Sidon sets; divisor sets; maximal independent sets; hypergraph transversals; sunflower; Pell equations; exact computation; integer sequences.

1 The question and the first surprise

A finite set $S \subset \mathbb{Z}$ is *Sidon* if

$$a + b = c + d, \quad a, b, c, d \in S,$$

forces equality of the unordered multisets $\{a, b\} = \{c, d\}$. Repeated summands are allowed, so $1 + 3 = 2 + 2$ is a genuine collision.

There are two different notions of being large. A Sidon subset is *maximum* if it has the largest possible cardinality. It is *maximal* if no further ambient element can be adjoined without destroying the Sidon property. A maximal set can be locally stuck well below the maximum.

For a finite set X , write

$$\text{Spec}_{\max}(X) = \{|S| : S \subseteq X \text{ is inclusion-maximal Sidon}\}.$$

We ask whether the possible maximal sizes fill every integer between their minimum and maximum. For background on Sidon sets, see O'Bryant's survey [3]. The distinction between

maximum and inclusion-maximal Sidon sets was studied explicitly by Dias da Silva and Nathanson [2]; here the focus is the full cardinality spectrum inside arithmetic divisor sets.

Our ambient sets are divisor sets

$$\mathcal{D}(n) = \{d \in \mathbb{Z}_{>0} : d \mid n\}.$$

The first squarefree level at which a gap can occur is four prime factors. The purpose of this paper is not merely to exhibit examples at that level, but to give a complete certificate-level classification of every possible four-prime gap mechanism. Two small examples already show that the phenomenon is not uniform:

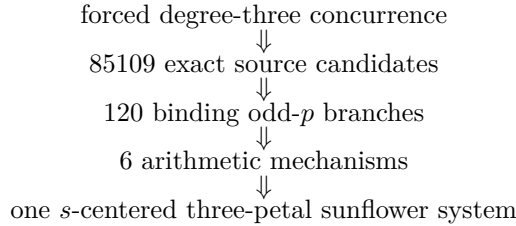
$$17255 = 5 \cdot 7 \cdot 17 \cdot 29, \quad \text{Spec}_{\max}(\mathcal{D}(17255)) = \{10, 11, 12, 14\},$$

where only 13 is missing, whereas

$$61509 = 3 \cdot 7 \cdot 29 \cdot 101, \quad \text{Spec}_{\max}(\mathcal{D}(61509)) = \{13, 15\}.$$

At first sight these look like different accidents. The main point of this paper is that they are born from the same local shape.

Main message. Every four-prime noninterval case comes from one of exactly six arithmetic mechanisms. Their generic branch systems are precisely the six arithmetically admissible orientations of a single s -centered three-petal sunflower. The finite computation is used only to prove that no other degree-three concurrence can produce a gap.



The associated increasing integer sequence has an independent life in the OEIS. Let A399802 denote the squarefree integers with exactly four prime factors whose maximal Sidon spectrum is not an interval [4]. The theorem below gives a complete classification of the *possible arithmetic mechanisms* underlying A399802: every term lies on one of six explicit arithmetic families, and no seventh family occurs. This is deliberately not an if-and-only-if parametrization of the sequence. A prime-valued point on one of the six families need not itself be a term, because additional collisions at a specialization can fill the generic gap.

Why four primes are the natural threshold

It is useful to place the four-prime problem in a larger hierarchy. For $k \geq 1$, let

$$E_k = \{n : n \text{ is squarefree, } \omega(n) = k, \text{ Spec}_{\max}(\mathcal{D}(n)) \text{ is not an interval}\}.$$

A finite ternary collision sieve proves that

$$E_1 = E_2 = E_3 = \emptyset;$$

see the companion treatment of the sharp four-prime threshold [5]. The set E_4 , listed increasingly, is A399802. The present theorem says exactly what can happen at this first nonempty level: every member must lie on one of six explicit arithmetic branches.

Above the threshold the situation changes again. If a new prime is sufficiently separated from the old divisor set, no collision can mix the old and new divisor layers. Consequently a four-prime gap can be lifted to five, six, and arbitrarily many prime factors. For example, starting from the spectrum $\{13, 15\}$, one separated prime produces

$$\{13, 15\} + \{13, 15\} = \{26, 28, 30\},$$

and a second separated prime produces

$$\{52, 54, 56, 58, 60\}.$$

So the role of four primes is more precise than merely “the first counterexample”: it is the first level at which a genuinely new gap mechanism can be born. Higher levels contain both inherited gaps and, potentially, new primitive mechanisms of their own.

2 From additive collisions to transversals

The right object is the collection of minimal additive obstructions.

Definition 2.1 (Collision clutter). Let $X \subset \mathbb{Z}$ be finite. A nontrivial collision is an equality

$$a + b = c + d$$

between two distinct unordered pairs from X . The *collision clutter* $\mathcal{C}(X)$ has vertex set X and, as edges, the inclusion-minimal supports of nontrivial collisions.

A collision support has three or four vertices. A four-point support containing a three-point collision is redundant and is discarded; this minimalization is important throughout.

Proposition 2.2 (Collision–transversal duality). *A subset $S \subseteq X$ is inclusion-maximal Sidon if and only if $X \setminus S$ is an inclusion-minimal transversal of $\mathcal{C}(X)$. Hence*

$$\text{Spec}_{\max}(X) = \{|X| - |T| : T \text{ is a minimal transversal of } \mathcal{C}(X)\}.$$

Proof. A subset is Sidon exactly when it contains no collision edge, so Sidon subsets are the independent sets of $\mathcal{C}(X)$. In any hypergraph, maximal independent sets are the complements of minimal transversals. $\square$

The translation matters because connected components now separate completely.

Lemma 2.3 (Component reduction). *If a clutter is the disjoint union of connected components $H_1, \dots, H_t$, then*

$$\text{TrSpec}_{\min}(H) = \text{TrSpec}_{\min}(H_1) + \dots + \text{TrSpec}_{\min}(H_t)$$

as a Minkowski sum. Consequently, if $\text{TrSpec}_{\min}(H)$ is not an interval, some connected component already has a noninterval transversal spectrum.

Proof. A minimal transversal is obtained by choosing a minimal transversal independently in each component. Cardinalities add, and a Minkowski sum of integer intervals is again an interval. $\square$

We shall also use the following general fact about sparse collision systems.

Proposition 2.4 (Degree-two interval theorem). *If every ground element of a finite clutter belongs to at most two edges, then the cardinalities of its minimal transversals form an integer interval.*

A short proof is recalled in section C. The consequence we need is immediate: a gap forces some divisor to lie in at least three minimal collision edges.

3 The one local obstruction: a three-petal sunflower

A clutter is a three-petal sunflower if its edges have the form

$$E_i = \{z\} \cup Q_i, \quad i = 1, 2, 3,$$

where the nonempty petals Q_i are pairwise disjoint. Its minimal transversals are especially simple: either take the center z , or avoid z and choose one element from each petal. Thus

$$\text{TrSpec}_{\min} = \{1, 3\}.$$

The missing transversal size 2 is the smallest possible spectral gap.

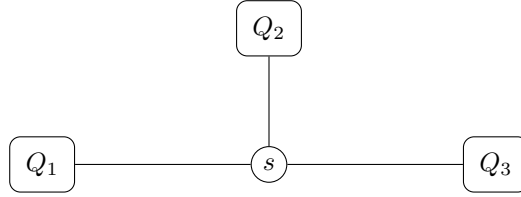


Figure 1: The local obstruction that drives the classification. The three minimal collision supports meet in the single divisor s and are otherwise disjoint. Their sizes are either 3, 4, 4 or 4, 4, 4.

For 61509, the mechanism can be seen completely at a glance.

From collisions to a sunflower. The three minimal collisions are

$$7 + 101 = 21 + 87, \quad 1 + 303 = 101 + 203, \quad 3 + 707 = 101 + 609.$$

Their supports are

$$\{7, 21, 87, 101\}, \quad \{1, 101, 203, 303\}, \quad \{3, 101, 609, 707\}.$$

They meet exactly in the common divisor $101 = s$, and deleting 101 leaves three pairwise disjoint petals. For this specialization these are the whole minimal collision clutter. Hence a minimal transversal either chooses the center, or one vertex from each petal:

$$\text{TrSpec}_{\min} = \{1, 3\}, \quad \text{Spec}_{\max}(\mathcal{D}(61509)) = \{16 - 1, 16 - 3\} = \{13, 15\}.$$

The example suggests that the sunflower should be the universal local obstruction, but that conclusion must not be assumed. A vertex of collision degree three need not lie in a sunflower in an arbitrary clutter, and non-sunflower three-edge concurrences also occur in the four-prime divisor lattice. The global step is therefore an exact arithmetic classification of *all* potentially gap-producing concurrences.

Proof roadmap

The proof follows one line from the first examples to the final classification.

Step 1. The examples above exhibit a local three-petal sunflower and show how a missing transversal size becomes a missing maximal-Sidon size.

Step 2. Collision–transversal duality turns the Sidon problem into a problem about minimal transversals. The degree-two interval theorem then shows that every gap forces a degree-three collision concurrence.

- Step 3.** An exact finite generator treats *every* such concurrence, without assuming sunflower geometry, and reduces the odd- p problem to a final interface of 120 exact branch ideals. These are mapped injectively to distinct identities in a canonical 897-branch reference ledger.
- Step 4.** The 120 current branches split as 7 s -centered, 80 center-1, 27 other-center, and 6 sunflower-free branches. Exact branch certificates account for all of them.
- Step 5.** Exactly six arithmetic systems survive: F1–F4 and Pell-A/B. Their equations are solved by hand.
- Step 6.** Inspecting those six systems reveals the promised bridge back to the local picture: their generic branch systems are the six orientations of one s -centered three-petal sunflower.

The computational part therefore supplies *completeness*, while the mathematical story remains the passage from a forced local concurrence to one universal sunflower geometry.

Two ledgers, two roles

The two recurring branch counts serve different purposes and should not be conflated.

- 120** Binding odd- p proof state after the source reduction. Every one of these branches is closed or identified as one of the six surviving families.
- 897** Canonical terminal reference ledger used for stable identities, provenance, and geometric classification. The full ledger is also replayed independently in the separate $p = 2$ case.

Thus the odd- p theorem does *not* require a fresh closure of all 897 reference identities. It requires the exact closure of the 120 source-image branches, together with their verified injection into the canonical ledger.

4 From degree-three concurrence to the final 120-branch interface

The degree-two theorem leaves one global completeness problem: classify every possible three-edge collision concurrence through a common divisor. This is where the exact finite computation enters. It generates the concurrence universe, imposes the corresponding polynomial relations, repeatedly recomputes the full generic minimal collision clutter, and closes a branch only by an exact arithmetic or finite-clutter certificate. No sunflower structure is assumed at this stage.

Theorem 4.1 (Exact finite reduction). *Let*

$$n = pqrs, \quad p < q < r < s$$

be a product of four distinct primes. If p is odd and $\text{Spec}_{\max}(\mathcal{D}(n))$ is not an interval, then its collision data lies on one of the 120 exact branches at the final source interface. Those 120 branches map injectively to 120 distinct identities in the canonical 897-branch reference ledger. The case $p = 2$ is handled separately by the exact terminal replay in theorem 5.1.

The proof is exhaustive rather than heuristic. Starting from all centered concurrences forced by theorem 2.4, the exact source reduction removes inconsistent, arithmetically impossible, and interval-filling branches until only 120 odd- p current branches remain. The historical endgame is indexed by 897 terminal identities, and exact ideal comparison identifies

each current branch with a distinct terminal identity. The number 897 is therefore a canonical reference frame for geometry and provenance; the binding odd- p closure is performed on the 120 source-image branches.

For the reader, only four facts about the computation are needed here:

- (i) every degree-three concurrence required by theorem 2.4 is generated;
- (ii) every imposed relation is handled with exact integer or polynomial arithmetic;
- (iii) the full generic minimal collision clutter is recomputed, so extra forced collisions that fill a putative gap are not missed;
- (iv) every branch is either given an exact exit certificate or passed forward exactly once to the final current interface.

The source-level replay semantics, provenance maps, certificate taxonomy, hard-gate discipline, checkpoints, and verifier contract are collected in section A. They are important for reproducibility, but not needed to follow the classification itself.

Numerically, the source reduction passes through 85109 candidates and reaches 120 current branches, which are then mapped into the canonical 897-identity ledger and composed to six surviving mechanisms. These are exact certificate milestones, not samples. The next section shows how the final current-branch arithmetic looks in practice.

5 The final 120-branch composition

Via the canonical terminal ledger, the 120 current odd- p branches fall into four geometric classes according to the center and support geometry of the degree-three core. Most are dispatched by a small repertoire of order, parity, divisibility, and forced-collision arguments. The following three branches are representative complete closures taken directly from the accepted current-branch proof path.

Representative branch closures

Sunflower-free branch 2183: a polynomial contradiction

The exact generator has already reconstructed the full generic clutter and placed branch 2183 in the sunflower-free class; what remains is its arithmetic certificate. The terminal relations give

$$r - q = q - p =: t.$$

Since p, q, r are odd, t is a positive even integer. A second terminal relation gives

$$q \mid s + 1, \quad s + 1 = nq$$

for some $n \geq 1$, and the same branch equations yield

$$nt = p - 1, \quad q \mid p^2 + 1.$$

Because $p = q - t$, the last divisibility becomes $q \mid t^2 + 1$. Moreover $p = nt + 1$ and $q = (n + 1)t + 1$, so $q > t$. Write

$$t^2 + 1 = hq.$$

Then $h < t$, while reduction modulo t gives $h \equiv 1 \pmod{t}$. Hence $h = 1$, and therefore

$$q = t^2 + 1, \quad p = t^2 - t + 1, \quad r = t^2 + t + 1.$$

Substitution into the final branch relation gives

$$(t^2 + 1)(t^3 - 3t^2 - 3) = 0.$$

The first factor is positive. The cubic equals -7 at $t = 2$ and 13 at $t = 4$, and its derivative is $3t(t - 2) > 0$ for $t > 2$. Hence it has no positive even integral zero, so branch 2183 is impossible.

Current center-1 branch 296: prime cancellation and factorization

Two exact branch relations, reduced modulo r , give residues whose difference is $(p + q)s$. Hence

$$r \mid (p + q)s.$$

Since r and s are distinct primes, $r \mid p + q$. The order $p < q < r$ gives

$$0 < p + q < 2r,$$

so necessarily $r = p + q$. A remaining branch equation is

$$pq - r - s + 1 = 0,$$

and therefore

$$s = pq - p - q + 1 = (p - 1)(q - 1).$$

For odd primes $p < q$, both factors exceed 1, contradicting the primality of s . Thus branch 296 is impossible.

Current other-center branch 1993: a large-prime cancellation

The exact current branch contains

$$p - 2q + r = 0, \quad s(r + 1) = p(q + r).$$

The second identity implies $s \mid p(q + r)$. Since $p < s$ and s is prime, $s \nmid p$, so

$$s \mid q + r.$$

But $0 < q + r < 2s$, hence $q + r = s$. In the odd- p endgame the primes q, r, s are odd, so the left side is even while the right side is odd, a contradiction. Thus branch 1993 is impossible.

Theorem 5.1 (Exact final composition). *For odd p , the final source interface consists of 120 branches and has the exact accounting*

$$120 = 7 + 80 + 27 + 6,$$

corresponding respectively to s -centered, center-1, other-center, and sunflower-free geometry in the canonical terminal ledger. Exactly six branches are the systems $F1, F2, F3, F4, \text{Pell-A}, \text{Pell-B}$. Of the remaining 114, exactly one is an ordered-prime interval exception and the other 113 are impossible. For $p = 2$, the canonical 897 terminal identities are closed independently by the exact decomposition $891 + 6 = 897$. Consequently no seventh gap mechanism survives.

Proof. For odd p , exact current-to-terminal identity comparison gives the class counts

$$7 \quad (s\text{-centered}), \quad 80 \quad (\text{center } 1), \quad 27 \quad (\text{other center}), \quad 6 \quad (\text{sunflower-free}).$$

The s -centered block contains the six family identities and one further branch that is closed exactly. All $34 = 1 + 27 + 6$ current non-center-1 nonfamily branches have explicit accepted branch certificates. The center-1 block is accounted for as

$$80 = 20 + 59 + 1 :$$

20 branches are closed by fresh elementary certificates, 59 imply exact systems in the independently certified UNIT-center impossibility ledger, and the remaining branch is terminal identity 708, corresponding to

$$(p, q, r, s) = (3, 5, 7, 17).$$

Its full collision clutter has interval maximal-Sidon spectrum $\{7, 8, 9, 10, 11\}$. Hence the odd- p current interface has six family exits and no unaccounted branch.

For $p = 2$, exact factorization of the same 897 terminal eliminants excludes 891 identities and leaves precisely the six residual IDs

$$164, 1564, 1623, 1635, 2173, 2507.$$

After fixing the admissible prime value of q in each row, exact Gröbner normal forms give the consequences

164	$q = 3, r = 4,$
1564	$q = 5, s = 2(r - 2),$
1623	$q = 5, r = 16,$
1635	$q = 3, s = 3r - 1,$
2173	$q = 3, 4s = 5r,$
2507	$q = 3, s = 2r - 3 = 3r - 2.$

Each contradicts primality or distinctness. Thus the even case contributes no mechanism either. □

Why the positive center-1 exception is harmless

One terminal center-1 system has the ordered-prime solution

$$(p, q, r, s) = (3, 5, 7, 17).$$

Its three defining relations are

$$pr + 1 = q + s, \quad pqr + pq = rs + 1, \quad ps + qr = qs + 1,$$

so at this specialization they give the 1-centered sunflower supports

$$\{1, 5, 17, 21\}, \quad \{1, 15, 105, 119\}, \quad \{1, 35, 51, 85\}.$$

If these were the whole clutter they would create the usual transversal gap. They are not. The full divisor set of $1785 = 3 \cdot 5 \cdot 7 \cdot 17$ has additional minimal collisions, including

$$\begin{aligned} 1 + 5 &= 2 \cdot 3, & 3 + 7 &= 2 \cdot 5, & 7 + 35 &= 2 \cdot 21, \\ 17 + 85 &= 2 \cdot 51, & 51 + 119 &= 2 \cdot 85, & 119 + 595 &= 2 \cdot 357. \end{aligned}$$

In fact the full minimal collision clutter has 27 edges, and exact minimal-transversal enumeration gives

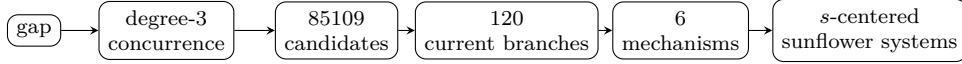
$$3z^5 + 61z^6 + 209z^7 + 56z^8 + z^9.$$

Thus

$$\text{TrSpec}_{\min} = \{5, 6, 7, 8, 9\}, \quad \text{Spec}_{\max}(\mathcal{D}(1785)) = \{7, 8, 9, 10, 11\}.$$

The extra collisions therefore fill the local sunflower gap rather than creating a seventh mechanism.

The proof funnel



The canonical 897-identity ledger supplies stable labels and geometric classes for the final comparison, but the binding odd- p closure is the exact composition of the 120 source-image branches. The role of the computer is completeness: it guarantees that every degree-three concurrence is either closed by an exact certificate or reaches this final interface. What survives is much more rigid than the size of the initial universe suggests: the six branches are six arithmetic realizations of one generic local configuration. Reproducibility details are deferred to section A.

6 Six orientations, six arithmetic families

After all eliminations, exactly six branches remain. Remarkably, their defining generic collision systems are the six arithmetically admissible orientations of a single s -centered three-petal sunflower. They are displayed in table 1; the equations are written in a scaled form so that the common divisor s is visible.

At the level of the full generic collision clutter, F2 and F4 are pure three-edge sunflowers. In F1, F3 and the two Pell branches the same sunflower core forces additional generic collisions. This is why one local obstruction can lead to several different global maximal-Sidon spectra without creating any new local mechanism.

branch	three sunflower collisions	petal type
<i>Polynomial branches</i>		
F1	$1 + pq = q + s; p + s = 2r; s + qrs = pqs + prs$	344
F2	$1 + ps = s + qr; p + qs = s + pqr; q + s = pq + pr$	444
F3	$1 + pr = q + s; p + qr = s + ps; s + rs = qs + pqs$	444
F4	$1 + qr = s + ps; p + pqr = s + qs; q + pr = pq + s$	444
<i>Pell branches</i>		
Pell-A	$1 + pqr = 2s; s + rs = ps + qs; s + qrs = pqs + prs$	344
Pell-B	$1 + s = 2pqr; s + rs = ps + qs; s + qrs = pqs + prs$	344

Table 1: The six arithmetically possible orientations of the s -centered sunflower. The type records the three support sizes.

The point of the table is not merely that these equations hold. They are already enough to derive the six parameter families by hand.

6.1 F1

From

$$1 + pq = q + s$$

and

$$p + s = 2r$$

we obtain

$$s = pq - q + 1, \quad r = \frac{p + s}{2}.$$

The key observation is that the third sunflower collision loses its common factor s and becomes linear in r . Dividing it by s gives

$$1 + qr = pq + pr,$$

so

$$r(q - p) = pq - 1.$$

Substitution yields

$$(p - 1)(q + 1)(p - q + 2) = 0.$$

Since $p > 1$ and $q > 0$,

$$q = p + 2.$$

Hence

$$q = p + 2, \quad r = \frac{p^2 + 2p - 1}{2}, \quad s = p^2 + p - 1.$$

6.2 F2

The first two equations give

$$qr = 1 + (p - 1)s, \quad pqr = p + (q - 1)s.$$

Here the first two collisions contain the same qr term at different scales; comparing them removes both r and s . Multiplying the first by p and comparing with the second gives

$$p(p - 1) = q - 1,$$

hence

$$q = p^2 - p + 1.$$

Solving the first and third equations for r and s now gives

$$\begin{aligned} q &= p^2 - p + 1, \\ r &= p^4 - 3p^3 + 4p^2 - 3p + 2, \\ s &= p^5 - 3p^4 + 5p^3 - 5p^2 + 4p - 1. \end{aligned}$$

At $p = 3$ this is $(3, 7, 29, 101)$, the example 61509.

6.3 F3

Dividing the third equation by s gives

$$1 + r = q + pq,$$

so

$$r = pq + q - 1.$$

The first equation gives

$$s = 1 + pr - q.$$

Thus the first and third collisions already determine r and s in terms of p, q ; the middle collision is the compatibility condition. Substituting these expressions in the second equation yields

$$(pq + q - 1)(p^2 + p - q - 1) = 0.$$

The first factor is positive, hence

$$q = p^2 + p - 1.$$

Therefore

$\begin{aligned} q &= p^2 + p - 1, \\ r &= p^3 + 2p^2 - 2, \\ s &= p^4 + 2p^3 - p^2 - 3p + 2. \end{aligned}$
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6.4 F4

The first equation is

$$1 + qr = (p + 1)s.$$

Again two collisions share the same scaled quantity $1 + qr$, so the parameter q is forced before r and s are solved. The left side of the second is $p(1 + qr)$, so comparison with

$$p + pqr = (q + 1)s$$

gives

$$q + 1 = p(p + 1).$$

Thus

$$q = p^2 + p - 1.$$

Solving the first and third equations for r, s gives

$\begin{aligned} q &= p^2 + p - 1, \\ r &= p^4 + p^3 - 2p^2 - p + 2, \\ s &= p^5 + p^4 - 3p^3 - p^2 + 4p - 1. \end{aligned}$
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6.5 The Pell pair

The last two sunflower equations are common to Pell-A and Pell-B. After division by s they read

$$1 + r = p + q,$$

$$1 + qr = pq + pr.$$

The first gives

$$r = p + q - 1.$$

Substitution into the second gives the single quadratic constraint

$$\boxed{p^2 + pq - q^2 - p + q - 1 = 0.}$$

Equivalently,

$$\boxed{(5p - 1)^2 - 5(2q - p - 1)^2 = 16.}$$

Thus the two apparently different branches share the same Pell spine. They differ only in the three-point petal:

$$\text{Pell-A:} \quad 1 + pqr = 2s, \quad \boxed{s = \frac{pqr + 1}{2}},$$

whereas

$$\text{Pell-B:} \quad 1 + s = 2pqr, \quad \boxed{s = 2pqr - 1}.$$

Remark 6.1 (The Pell spine). The Pell structure is not accidental: it is what remains after eliminating the two common sunflower equations. Putting $X = 5p - 1$ and $Y = 2q - p - 1$ turns the common constraint into

$$X^2 - 5Y^2 = 16.$$

This is a generalized Pell equation. Its integral points form finitely many orbits under the norm-one unit action, and in particular the conic has infinitely many integral points. The four-prime problem imposes additional congruence, ordering and simultaneous-primality conditions on p, q, r, s . No infinitude of prime-valued specializations of either Pell branch is claimed here.

7 Classification and the sunflower bridge

Combining the exhaustive source reduction, the exact final composition on the current 120-branch interface, and the six elementary derivations gives the main result.

Theorem 7.1 (Six-family classification). *Let*

$$n = pqrs, \quad p < q < r < s$$

be a product of four distinct primes. If $\text{Spec}_{\max}(\mathcal{D}(n))$ is not an interval, then (p, q, r, s) belongs to one of the six families $F1, F2, F3, F4, \text{Pell-A}, \text{Pell-B}$ displayed above. There is no seventh arithmetic mechanism.

Here “classification” refers to the necessary arithmetic mechanisms. The theorem does not assert that every prime-valued specialization of one of the six families has a noninterval spectrum.

Corollary 7.2 (Sequence interpretation). *Every term of A399802 is obtained from one of $F1, F2, F3, F4, \text{Pell-A}$, or Pell-B . No seventh arithmetic family occurs.*

The assertion is one-way at the level of algebraic parametrizations. Exceptional prime-valued specializations can acquire additional collision edges and therefore fail to be terms of A399802; this occurs already for $(3, 5, 7, 11)$ in $F1$ and $(3, 5, 7, 53)$ in Pell-A .

The sunflower statement now follows without any further classification.

Corollary 7.3 (Sunflower-system bridge). *If $\text{Spec}_{\max}(\mathcal{D}(pqrs))$ is not an interval, then its collision equations satisfy one of the six s -centered three-petal sunflower systems in table 1. Thus every surviving arithmetic mechanism has one of the six orientations of the same generic branch system.*

Proof. By theorems 4.1 and 5.1, one of the six systems in table 1 holds. In every row the three displayed generic collision supports all contain s , and after removing s their petals are pairwise disjoint. Hence the six surviving branch systems are exactly six orientations of one s -centered three-petal sunflower. $\square$

The phrase “generic branch system” is essential. A special prime-valued parameter may create additional collisions, and a newly created smaller support can make one of the displayed four-point generic supports nonminimal in the exact specialized clutter. The six-family theorem and the sunflower-system bridge are unaffected, because they concern the exact arithmetic identities defining the surviving branch. No claim is made here that the same three displayed supports remain minimal edges at every specialization.

This is a classification of *candidate mechanisms*, not a statement that every prime-valued specialization is automatically a term of the noninterval sequence. A specialization can create additional collisions. Those additional edges can change the minimal-transversal spectrum and can even fill the gap.

8 One local mechanism, several global spectra

The sunflower is the common generic branch system, but it is not always the whole collision clutter. The generic local core is therefore the same in all six cases; the different spectra arise because the surrounding generic collision clutter differs, and specialized clutters may acquire further minimal edges.

For a branch, call an edge *generic* if its collision equality holds identically on the algebraic family. The generic minimal collision clutters have the transversal data in table 2. The counts are obtained by direct minimal-transversal enumeration of the displayed generic clutters in section B.

branch	generic minimal edges	$\text{TrSpec}_{\min}$	generic $\text{Spec}_{\max}$
F1	7	$\{2, 4, 5, 6\}$	$\{10, 11, 12, 14\}$
F2	3	$\{1, 3\}$	$\{13, 15\}$
F3	5	$\{2, 4\}$	$\{12, 14\}$
F4	3	$\{1, 3\}$	$\{13, 15\}$
Pell-A	5	$\{2, 3, 5\}$	$\{11, 13, 14\}$
Pell-B	5	$\{2, 3, 5\}$	$\{11, 13, 14\}$

Table 2: The generic full collision clutters. F2 and F4 are pure three-edge sunflowers. In the other four branches, further collision edges are forced identically.

The corresponding minimal-transversal generating polynomials are

$$M_{F1}(z) = z^2 + 45z^4 + 27z^5 + 6z^6,$$

$$M_{F2}(z) = M_{F4}(z) = z + 27z^3,$$

$$M_{F3}(z) = 4z^2 + 72z^4,$$

and

$$M_{\text{Pell}}(z) = z^2 + 18z^3 + 81z^5.$$

This gives a useful conceptual picture: the same generic sunflower system can sit inside several different globally forced collision systems. The full clutter, not the sunflower system alone, decides which cardinalities are ultimately missing.

9 Exceptional specializations and genuine examples

The distinction between a generic branch system and an exact specialized clutter is essential.

The small F1 specialization

$$(p, q, r, s) = (3, 5, 7, 11)$$

has additional collisions and an interval spectrum. At $p = 5$, however, F1 gives

$$(5, 7, 17, 29), \quad n = 17255,$$

with

$$\text{Spec}_{\max}(\mathcal{D}(17255)) = \{10, 11, 12, 14\}.$$

For Pell-A, the prime-valued specialization

$$(p, q, r, s) = (3, 5, 7, 53)$$

is also exceptional. At this specialization the generic support $\{s, ps, qs, rs\}$ is no longer minimal because additional collisions appear; this is a concrete reason to distinguish the generic sunflower system from the exact specialized minimal clutter. The resulting spectrum is nevertheless easy to compute exactly:

$$\text{Spec}_{\max}(\mathcal{D}(3 \cdot 5 \cdot 7 \cdot 53)) = \{9, 10, 11, 12\},$$

so it does not belong to the noninterval sequence.

By contrast, Pell-B has the prime specialization

$$(p, q, r, s) = (883, 1429, 2311, 5832071953),$$

whose maximal Sidon spectrum is

$$\{11, 13, 14\}.$$

It gives

$$17006531635400653081,$$

the eighth term of the resulting increasing sequence.

F3 also occurs genuinely. The prime tuple

$$(463, 214831, 99681583, 46152358099)$$

has spectrum

$$\{12, 14\}.$$

The examples above show that several of the non-F1/F2 mechanisms are arithmetically genuine rather than formal artifacts of elimination. The present classification is nevertheless not a theorem on simultaneous prime values of the defining polynomials. In particular, we do not know—and do not claim—that any of the six parameter families contains infinitely many prime-valued specializations. For each specialization the full collision clutter must also be checked, since extra collisions can fill the generic gap.

Accordingly, A399802 begins

$$\begin{aligned} &17255, 61509, 1330043, 2673062363, 250553227511723, \\ &4851462337681343, 235118392968359123, 17006531635400653081, \dots \end{aligned}$$

The six-family theorem therefore gives a complete classification of the possible arithmetic mechanisms underlying this sequence, rather than merely a method for producing examples.

10 The threshold hierarchy beyond four primes

The classification above concerns the first level at which a squarefree divisor set can have a gap. Once such a gap exists, however, a very simple separation principle carries it to all higher levels. We include the argument because it clarifies what should count as genuinely new beyond four primes.

Lemma 10.1 (Collision separation). *Let $\ell \nmid m$ be prime and suppose $\ell > 2m$. Every nontrivial Sidon collision in $\mathcal{D}(m\ell^e)$ lies entirely in one ℓ -adic layer*

$$\ell^j \mathcal{D}(m), \quad 0 \leq j \leq e.$$

Proof. Write every divisor as $\ell^j a$ with $a \mid m$, and consider a collision

$$x + y = u + v.$$

Divide by the smallest power of ℓ occurring among the four terms. Reducing modulo ℓ , the contribution from the lowest layer on either side is a sum of zero, one, or two positive divisors of m , hence lies in $[0, 2m]$. Since $2m < \ell$, congruence modulo ℓ is actual equality.

If one side has a lowest-layer term and the other has none, equality is impossible. If one side has two lowest-layer terms and the other has one, equality of the low-layer coefficients would leave the latter side with one positive higher-layer term while the former has none, which is impossible. If each side has exactly one lowest-layer term, those two terms are equal and cancel, after which the remaining two terms are equal as well; the collision is trivial. Hence every nontrivial collision has two lowest-layer terms on each side, so all four terms lie in the same layer. $\square$

Theorem 10.2 (Separated-prime lifting). *Let $\ell \nmid m$ be prime with $\ell > 2m$. Then for every $e \geq 0$,*

$$\text{Spec}_{\max}(\mathcal{D}(m\ell^e)) = \underbrace{\text{Spec}_{\max}(\mathcal{D}(m)) + \cdots + \text{Spec}_{\max}(\mathcal{D}(m))}_{e+1 \text{ Minkowski summands}}.$$

Proof. By theorem 10.1, the collision clutter of $\mathcal{D}(m\ell^e)$ is the disjoint union of $e + 1$ scaled copies of the collision clutter of $\mathcal{D}(m)$. Maximal independent sets are chosen independently in the components, so their cardinalities add. $\square$

Starting with

$$A = \text{Spec}_{\max}(\mathcal{D}(61509)) = \{13, 15\},$$

one squarefree lift replaces A by

$$A + A = \{26, 28, 30\},$$

and the next replaces it by

$$A + A + A + A = \{52, 54, 56, 58, 60\}.$$

More generally, after $k - 4$ squarefree separated-prime lifts, the original spectrum has been added to itself

$$L = 2^{k-4}$$

times, and therefore

$$\underbrace{A + \cdots + A}_{L \text{ summands}} = \{13L, 13L + 2, \dots, 15L\}.$$

For $k \geq 5$ the integer L is even, so exactly the L odd integers in the convex hull are missing. Since at every stage the next prime may be chosen arbitrarily large, this yields infinitely many squarefree examples for each fixed $k \geq 5$.

Together with the ternary contiguity theorem [5], this gives the sharp hierarchy

k	squarefree maximal Sidon spectra
1, 2, 3	always intervals,
4	first possible gaps; all gap mechanisms classified here,
$k \geq 5$	infinitely many gaps by separated-prime lifting.

This suggests a natural notion for the next stage. Call a noninterval squarefree divisor set *primitive with respect to separated-prime lifting* if it cannot be obtained from a smaller noninterval divisor set by repeatedly adjoining primes larger than twice the current integer. In this language, the present paper classifies the first possible primitive level. The next questions are then not whether gaps exist for five or six prime factors—they certainly do—but whether genuinely new primitive gap cores appear there, and if so how they should be classified.

Concluding perspective

The four-prime classification has two complementary outputs. Combinatorially, it identifies a single generic local system—an s -centered three-petal sunflower system—behind every gap at the first nontrivial squarefree level, while the full collision clutter explains the different global spectra and specialization effects. Arithmetically, it converts that obstruction into six explicit families and hence into a structural classification of A399802. The exact finite certificate is the bridge between these two descriptions: it proves that no additional support geometry or arithmetic family has been omitted.

11 Outlook: primitive gap cores

Beyond four prime factors, separated-prime lifting already supplies infinitely many inherited gaps. The next structural question is therefore narrower: do genuinely new primitive gap cores occur at five prime coordinates, and can they again be classified by finitely many local collision templates? Any such core must use all five coordinates essentially rather than split as a four-prime obstruction plus a distant layer.

The four-prime result suggests that the right objects are full generic collision clutters together with their minimal-transversal spectra, not isolated collision triples. Determining whether the primitive five-prime part is finite up to natural arithmetic parametrization is the most direct continuation of the present work.

A Hybrid certificate, current-branch composition, and the 897-reference ledger

This appendix states the proof contract for the computer-assisted part. It deliberately avoids the obsolete residual normal forms sometimes used in working notes; in particular, no R1/R2/R3 architecture and no general “full-support principle” is part of the proof.

Reader’s map of the computational evidence. There are two reproducibility layers, with different logical roles.

Layer	What it establishes	What it does not claim
Source reconstruction record	Generation of the concurrence universe, specialization checks, and the exact source-wide funnel ending at the 120 current branches.	It is not needed for a quick rerun of the already frozen final composition.
Minimal final replay	Re-executes the UNIT endgame, the current non-center-1 and center-1 closures, the complete $p = 2$ terminal replay, and the final 120-branch composition gate.	It does not claim to regenerate the early $1059 \rightarrow \dots \rightarrow 120$ source funnel in the same run.

The first layer establishes source-level completeness; the second is the compact independently rerunnable final proof object.

A.1 Reproducibility architecture

The archived 897 identities are a common reference frame for geometry and provenance. The binding odd- p proof path is the source reduction to the final 120 current branches followed by exact branchwise composition. A current branch is identified by its exact polynomial ideal together with its imposed factor choice; integer labels are only ledger indices. Historical membership may be used for provenance and post-hoc comparison, never as a closure criterion.

The front end independently regenerates

$$1059 \rightarrow 366 \rightarrow 52150 \rightarrow 17788 \rightarrow 115251 \rightarrow 92800 \rightarrow 85109 \rightarrow 68449 \rightarrow 3842,$$

with the exact split $3842 = 3803 + 39$. The 39 complementary cases are independently certified impossible. The corrected arithmetic replay then proceeds source-wide as

$$\begin{aligned} 3803 &\rightarrow 2541 \rightarrow 1545 \rightarrow 1039 \rightarrow 528 \rightarrow 500 \rightarrow 411 \\ &\rightarrow 395 \rightarrow 363 \rightarrow 335 \rightarrow 150 \rightarrow 144 \rightarrow 141 \rightarrow 120. \end{aligned}$$

The final 120 current branches are mapped bijectively, by exact branch algebra and identity checks, to 120 distinct identities in the canonical 897 terminal reference ledger. The terminal algebra has been independently recovered for all 897 identities, and recomputation of their full generic collision clutters reproduces the geometric partition

$$897 = 46 + 603 + 227 + 21.$$

Within the s -centered block, exact ideal comparison identifies precisely the six family rows F1–F4 and Pell-A/B. The center-1 interval exception $(3, 5, 7, 17)$ is independently identified and its full-clutter transversal polynomial is recomputed exactly. The other-center census is 43, 60, 54, 6, 17, 22, 16, 6, 3 for centers $p, q, r, pq, pr, qr, ps, qs, rs$, respectively. All 21 sunflower-free rows are independently confirmed to be sunflower-free; their degree-three cores form seven exact prime-coordinate orbits. For $p = 2$, the terminal replay is complete: exact factorization of $F(2, q)$ excludes 891 identities and leaves exactly six residual rows, whose stated consequences are verified by exact Gröbner normal forms.

This distinction is deliberate: the canonical 897 identities organize comparison and geometry, while the source replay is allowed to certify branches earlier by stronger exact arguments. The final fail-closed composition gate, B2B69 v9, closes the current-branch proof path. It freshly replays the complete $p = 2$ terminal block as $891 + 6 = 897$, verifies all 34 current non-center-1 nonfamily branches by explicit accepted certificate rows, and partitions the 80 current center-1 branches into 20 fresh elementary closures, 59 exact UNIT-center

impossibilities, and the unique interval exception 708, corresponding to $(3, 5, 7, 17)$. Together with the six family exits this accounts for all 120 current terminal branches with residual zero. The UNIT-center input is independently certified by the accepted replay $1275 \rightarrow 63 \rightarrow 0$. The failed historical B2B43 membership closure is not used. Checkpoints are accelerators only; no checkpoint count is a mathematical premise. The archive README distinguishes regenerated source-stage material from frozen checksum-linked replay inputs.

A.2 The finite universe and completeness of the generator

Label the 16 divisors of $n = pqrs$ by bit vectors

$$u = (u_1, u_2, u_3, u_4) \in \{0, 1\}^4, \quad m_u = p^{u_1} q^{u_2} r^{u_3} s^{u_4}.$$

There are

$$\binom{16+1}{2} = 136$$

unordered divisor pairs when repeated summands are allowed. For two distinct unordered pairs $\pi = \{u, v\}$ and $\rho = \{x, y\}$, put

$$F_{\pi, \rho} = m_u + m_v - m_x - m_y \in \mathbb{Z}[p, q, r, s].$$

A collision is exactly $F_{\pi, \rho} = 0$. Its support has three or four vertices. Whenever a four-point support contains an active three-point collision, the four-point support is deleted because it is not minimal.

By theorem 2.4, a gap forces some divisor to occur in at least three minimal collision edges. Hence it is enough to generate every triple of minimal collision supports through a common vertex. No sunflower assumption is made. The only identifications permitted by the generator are literal duplicates obtained by exchanging the two sides of an equality, exchanging the two summands on one side, or permuting the three equations.

The main text displays only the high-level proof funnel; the source-level counts and replay status are recorded in the preceding subsection. At every imposed polynomial relation the full generic minimal collision clutter is recomputed before a branch is called dangerous. Thus extra forced collisions that fill a putative gap are incorporated rather than ignored.

Software, branch identity, and replay semantics. All symbolic computations are carried out in Maple 2026 with exact integer and rational-polynomial arithmetic. No branch decision uses floating-point arithmetic, a bounded prime search, or probabilistic primality testing. Each stage either regenerates its input from an earlier source stage or reads a checkpoint that is reproducible from source. A branch is identified mathematically by its exact polynomial ideal together with the imposed factor branch, not by an incidental integer label. When reconstructed and archived numbering differ, explicit identity maps are obtained from exact Gröbner-basis equality. The invariant is

every generated branch is either certified closed or passed forward exactly once.

A stronger exact sieve may therefore close an archived identity earlier than the historical computation did; the branchwise certificate, not membership in an old stage list, is the proof object.

A.3 Canonical terminal geometry and the current odd- p composition

Recomputation of the full generic clutters of the canonical 897 identities gives the exact geometric partition

$$897 = 46 + 603 + 227 + 21,$$

corresponding to s -centered, center-1, other-center sunflower, and sunflower-free geometry. This partition is used for branch identity and structural interpretation; it is not, by itself, an odd- p closure theorem.

The binding odd- p composition starts after the source-wide reduction. The final 120 current branches map injectively to 120 distinct terminal identities and split as

$$120 = 7 + 80 + 27 + 6.$$

The seven s -centered current branches contain exactly the six family identities F1–F4 and Pell-A/B plus one exact exclusion. The $34 = 1 + 27 + 6$ current non-center-1 nonfamily branches are all closed by explicit accepted certificates. On the center-1 block, 20 current branches are closed by fresh elementary certificates, 59 imply exact systems in the independently certified UNIT-center impossibility ledger, and terminal branch 708 is the unique interval exception. Thus

$$80 = 20 + 59 + 1$$

with no residual branch. The exact full-clutter computation for branch 708 gives the tuple $(3, 5, 7, 17)$, 27 minimal collision supports, and minimal-transversal polynomial

$$3z^5 + 61z^6 + 209z^7 + 56z^8 + z^9,$$

hence the interval maximal-Sidon spectrum $\{7, 8, 9, 10, 11\}$.

The terminal geometry remains useful beyond the current image: the full canonical ledger has 46 s -centered identities, 603 center-1 identities, 227 other-center sunflower identities, and 21 sunflower-free identities, with seven exact prime-coordinate orbits among the latter. These counts are provenance and geometry data; no closure decision in the final proof depends on membership in those historical classes.

A.4 The even case

For $p = 2$ the same terminal list is specialized exactly. Exact factorization of the 897 eliminants $F(2, q)$ yields 891 rows with no admissible prime root $q > 2$ and exactly the six residual IDs 164, 1564, 1623, 1635, 2173, 2507. No bound on q is used. For each residual, the unique admissible prime root ($q = 3$ or $q = 5$) is fixed first by the univariate factorization; after that specialization, exact Gröbner normal forms verify the consequences displayed in the proof of theorem 5.1. Thus the even terminal block is replayed exactly as $891 + 6 = 897$.

A.5 Certificate types and specialization soundness

The finite reduction uses a small collection of exact certificate types. We record their logical meaning because the stage counts alone are not proof objects.

Ideal certificate. An exact Gröbner-basis computation either proves polynomial inconsistency or identifies two branch descriptions as the same ideal. No numerical root approximation is used.

Ordered-cone sign certificate. After an exact substitution by positive gap variables encoding $p < q < r < s$ (and, where needed, the relevant parity cone), a polynomial with coefficients of one strict sign cannot vanish on the admissible cone.

Modulus certificate. If the branch equations force a modulus $g > 0$ to divide an exact polynomial residue R , while the ordered cone proves $0 < |R| < g$, the branch is impossible. The moduli used by the replay are themselves exact polynomial quantities such as p , $p - 1$, $q - p$, $r - q$, or $s - r$.

Resultant certificate. If two branch polynomials vanish, their exact resultant in the eliminated variable must vanish. Hence a sign or modulus certificate proving that resultant nonzero on every admissible parity cone is an exact impossibility certificate for the branch.

Finite-clutter cert. Once a full minimal collision clutter is fixed, its minimal transversals are enumerated exactly on the 16 divisor vertices. The resulting transversal polynomial and size spectrum are therefore finite combinatorial certificates, not sampled data.

Proposition A.1 (Soundness of the computational certificate types). *Every closure decision justified by the certificate types above is sound: the branch is either incompatible with an ordered-prime realization or has an exactly certified interval full-clutter spectrum. An ideal-equality certificate used only for deduplication merely identifies equivalent branch descriptions and is not itself a closure.*

Proof. An ideal inconsistency certificate is an identity in an exact polynomial ring; an ideal-equality certificate only merges equivalent descriptions. An ordered-cone sign certificate proves that a required polynomial cannot vanish on the admissible positive cone. A modulus certificate combines exact divisibility with the strict inequality $0 < |R| < g$. A resultant certificate uses the necessary vanishing of an exact elimination resultant and then reduces to a sign or modulus contradiction. Finally, a finite-clutter certificate enumerates all minimal transversals of a fixed finite clutter, so its spectrum is exact. None of these implications uses numerical approximation or a bounded search over primes. $\square$

There is one important specialization issue. Calling a *generic* clutter interval is not by itself a certificate that every specialization on the same algebraic branch is interval: a specialization may create additional minimal collision edges. Accordingly, a generic-interval exit is logically valid only after its exceptional collision loci have been split into exact subbranches and each subbranch has been proved impossible or interval, or routed to an independently certified later branch. These specialization checks belong to the source-stage reduction that precedes the final 120-branch interface; the minimal final composition replay does not substitute for them.

A.6 Verifier contract

The full proof contract consists of the following finite assertions using exact arithmetic. The first items describe the source-stage reconstruction; the final items describe the frozen current-branch composition.

- (C1) Generate the 16 divisor monomials and all 136 unordered pairs, including repetitions.
- (C2) Generate every nontrivial pair-sum equality and reduce it to its inclusion-minimal support.
- (C3) Generate every three-edge concurrence forced by the degree-two theorem, without an unproved divisor-lattice symmetry reduction.

- (C4) Recompute the full generic minimal collision clutter whenever a new polynomial relation is imposed.
- (C5) Reproduce the corrected finite accounting. In particular, regenerate

$$1059 \rightarrow 366 \rightarrow 52150 \rightarrow 17788 \rightarrow 115251 \rightarrow 92800 \rightarrow 85109 \rightarrow 68449 \rightarrow 3842,$$

verify the exact split $3842 = 3803 + 39$, certify the 39 complement cases arithmetically impossible, and reproduce the corrected source-wide arithmetic chain

$$\begin{aligned} 3803 &\rightarrow 2541 \rightarrow 1545 \rightarrow 1039 \rightarrow 528 \rightarrow 500 \rightarrow 411 \\ &\rightarrow 395 \rightarrow 363 \rightarrow 335 \rightarrow 150 \rightarrow 144 \rightarrow 141 \rightarrow 120. \end{aligned}$$

Verify that the final 120 current branches map bijectively to 120 distinct identities in the canonical 897 terminal ledger.

- (C6) Verify the provenance map from the reconstructed branch ideals to the archived identities used by the terminal ledger. No closure decision may depend on membership in an archived stage list; archived membership may be used only for post-hoc identity and accounting checks.
- (C7) Discharge every specialization watchlist created by a generic-interval exit. In particular, for each exceptional locus on which an additional minimal collision appears, verify an exact impossibility or interval certificate or an explicit routing to another certified branch.
- (C8) Recover the canonical terminal algebra and geometry needed for provenance: the partition $46 + 603 + 227 + 21$, the six s -centered family ideals, and the seven sunflower-free degree-three-core orbits.
- (C9) On the final odd- p current interface, verify the exact class split $7 + 80 + 27 + 6 = 120$, exactly six family exits, all 34 non-center-1 nonfamily closures, and the center-1 accounting $20 + 59 + 1 = 80$, with the last term equal to branch 708 and no unaccounted current branch.
- (C10) Verify that branch 708 corresponds to $(3, 5, 7, 17)$ and has exact full-clutter interval spectrum $\{7, 8, 9, 10, 11\}$.
- (C11) For $p = 2$, verify the exact $891 + 6 = 897$ decomposition: factor every $F(2, q)$ exactly, obtain precisely the six residual IDs 164, 1564, 1623, 1635, 2173, 2507, and verify their terminal consequences by exact specialization and Gröbner normal form.
- (C12) Return no unaccounted current branch and no unresolved even-case terminal identity.

All arithmetic in this contract is over integers or exact polynomial rings. There is no floating-point decision, probabilistic primality assumption, or prime cutoff. The computational output needed for the theorem is therefore

$$F1, F2, F3, F4, \text{ Pell-A, Pell-B; no unaccounted branch.}$$

Together with the hand derivations in section 6, this proves theorem 7.1.

A.7 Fail-closed replay discipline

For reproducibility, the implementation follows a fail-closed convention. Every hard consistency check is executed inside the stage procedure, and a stage prints its success marker only after all required counts, exact identities, certificate checks, specialization-watchlist checks, and post-hoc provenance comparisons have passed. A failed assertion therefore cannot be followed by a misleading success line from later top-level commands.

Historical ledgers are permitted as *targets for comparison* and as provenance dictionaries, but not as inputs to a mathematical closure decision. In particular, a regenerated stage must decide from its current exact polynomial data whether a branch closes; only afterward may the resulting set be compared with an archived stage boundary. This prevents a stale count or historical stage list from being silently promoted to a proof assumption.

The archive follows the two-layer distinction in the reader's map above. Historical ledgers and checkpoints are provenance aids or accelerators, not mathematical premises; the final replay uses only frozen exact inputs whose source provenance is separately recorded.

Clean-directory replay validation. The final v16 three-file referee distribution was reconstructed in a fresh Windows directory using only the paper, the human-readable bundle, and its START-HERE instructions, and was replayed with Maple 2026. The acceptance test combines byte-exact SHA-256 checks for outputs whose bytes are stable with canonical-output checks for three reports whose raw text may differ only by an absolute source path or by Maple's algebraically equivalent term-printing order. The run returns **FINAL REPLAY: PASS**. The package also supplies an **EXPECTED_FILES.txt** inventory giving the expected core filenames, byte sizes, and SHA-256 hashes; the manifest files themselves are excluded from self-reference.

B The generic full collision clutters

For reference we list the generic minimal supports. These are the supports whose collision equalities hold identically on the corresponding algebraic branch. In a specialized clutter additional supports may appear.

For F1 the seven generic edges are

$$\begin{aligned} &\{p, r, s\}, \quad \{pq, qr, qs\}, \quad \{1, q, s, pq\}, \quad \{1, pq, pr, qr\}, \\ &\{1, pr, qr, qs\}, \quad \{r, qr, rs, pqr\}, \quad \{s, pqs, prs, qrs\}. \end{aligned}$$

Their minimal-transversal generating polynomial is

$$z^2 + 45z^4 + 27z^5 + 6z^6.$$

For F2 and F4 the support clutter is the pure sunflower

$$\{1, s, ps, qr\}, \quad \{p, s, qs, pqr\}, \quad \{q, s, pq, pr\},$$

with generating polynomial

$$z + 27z^3.$$

The additive orientations of these three supports differ between F2 and F4.

For F3 the five generic edges are

$$\{1, q, r, pq\}, \quad \{1, q, s, pr\}, \quad \{p, s, ps, qr\}, \quad \{r, s, pq, pr\}, \quad \{s, qs, rs, pqs\}.$$

Their generating polynomial is

$$4z^2 + 72z^4.$$

For both Pell branches the five generic supports are

$$\{1, s, pqr\}, \quad \{1, p, q, r\}, \quad \{1, pq, pr, qr\}, \quad \{s, ps, qs, rs\}, \quad \{s, pqs, prs, qrs\}.$$

The first support has opposite three-point orientation in Pell-A and Pell-B, but the generic support clutter is the same. Its minimal-transversal generating polynomial is

$$z^2 + 18z^3 + 81z^5.$$

C Why degree two cannot create a gap

We give the reduction behind theorem 2.4. Let a finite clutter H have edges $E_1, \dots, E_m$, and suppose every ground element belongs to at most two edges. Two ground elements have the same *incidence type* if they lie in exactly the same clutter edges. Compress each nonempty incidence type to one representative. A degree-two type shared by E_i, E_j becomes an ordinary edge ij of an incidence graph G on vertices $E_1, \dots, E_m$; a clutter edge E_i containing a private ground element is marked.

Lemma C.1 (Incidence compression). *A minimal transversal contains at most one ground element from each incidence type.*

Proof. Two elements in the same incidence type hit exactly the same clutter edges. If a transversal contained both, deleting either one would leave a transversal, contradicting minimality. $\square$

A minimal transversal therefore becomes an inclusion-minimal cover of $V(G)$ by ordinary graph edges together with allowed private “halfedges” at marked vertices.

Lemma C.2 (Minimal covers are star forests). *In an inclusion-minimal such cover, the selected ordinary edges form a disjoint union of stars, and no vertex incident with a selected ordinary edge carries a selected private halfedge.*

Proof. Every selected ordinary edge must have an endpoint not covered by any other selected object; otherwise that edge could be deleted. Hence every selected edge has an endpoint of selected degree one. A connected graph in which every edge has a degree-one endpoint is a star. A selected private halfedge at a vertex already incident with a selected ordinary edge would be redundant. $\square$

Thus every minimal transversal corresponds to a partition of the m clutter edges into nontrivial stars of G and marked singletons. If there are c nontrivial star blocks, then the transversal has size

$$m - c,$$

because a star on t collision vertices uses $t - 1$ shared ground elements while a marked singleton uses one private ground element.

To remove the marked singletons, attach to every marked vertex v two new vertices a_v, b_v and the edges va_v, a_vb_v . Call the augmented graph $\hat{G}$.

Lemma C.3 (Tail correspondence). *Partitions of $V(G)$ into c nontrivial stars and marked singletons are in bijection with ordinary star partitions of $\hat{G}$ having $c + |P|$ blocks, where P is the set of marked vertices.*

Proof. For a marked vertex v , the leaf b_v forces a_v into the same star-partition block. That block is either $\{a_v, b_v\}$, leaving v in a nontrivial star inherited from G , or $\{v, a_v, b_v\}$, which represents a marked singleton. The choices are independent over the marked vertices, and each attached tail contributes exactly one block. $\square$

The graph $\hat{G}$ has no isolated vertices. Andreatta, De Francesco, De Giovanni and Serafini proved that the possible cardinalities of star partitions of a graph without isolated vertices form an integer interval [1, Theorem 6]. By the tail correspondence, c therefore runs through an integer interval, and the minimal-transversal size $m - c$ runs through an integer interval as well. This proves theorem 2.4.

Supplementary Materials

The supporting computational archive separates the source reconstruction record from the minimal final replay. It contains the exact branch and certificate records needed by the final composition, explicit current-to-terminal provenance maps, the accepted UNIT-center replay, the current noncenter and center-1 closure sources, the authoritative B2B69 v9 final composition gate, recorded successful outputs, SHA-256 checksum manifests, and an expected-file inventory with byte sizes and hashes. A clean-directory reconstruction of the v16 three-file referee distribution under Windows 11 and Maple 2026 returns **FINAL REPLAY: PASS**. Portable validation uses byte-exact hashes where appropriate and canonical-output equivalence where absolute paths or algebraically equivalent Maple print order can change raw bytes. The frozen final replay records zero unaccounted current branches and does not use B2B43 membership as a closure criterion. The archive is publicly available on Zenodo at doi:10.5281/zenodo.22720962.

Author Contributions

The author is solely responsible for the conception of the project, the mathematical results, the verification of the proofs and computational certificates, and the final submitted manuscript.

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Data Availability Statement

No empirical dataset is used. The computational archive accompanying the submitted manuscript contains the available source-stage reconstruction material, exact certificate and ledger files used by the final replay, the accepted final verifier sources, successful replay outputs, SHA-256 checksum manifests, and a filename/size/hash inventory for the core replay package. Its README distinguishes the recorded source reconstruction from the smaller final composition replay and documents the successful clean-directory three-file replay. The corresponding public archive is the Zenodo record cited above.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used generative artificial-intelligence tools based on large language models (LLMs) for mathematical exploration, development and checking of proof arguments, organization and auditing of finite case analyses, manuscript structuring, substantive drafting and rewriting, and language editing. The author independently reviewed, edited, and verified the mathematical statements, proofs, computations, references, and wording and takes full responsibility for the integrity and accuracy of the publication.

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